

IoT-BASED CLIMATE CHANGE PREDICTION SYSTEM

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ABSTRACT

Climate change is one of the most significant challenges to every country's development, ravaging havoc on the lives of all people on this planet. Researchers have raised numerous research and studies of strategies for tracking climate change. The current climate change tracking method in Rwanda employs a weather station model, in which numerous fixed weather stations are installed throughout the country; however, due to its immobility, this process cannot cover the entire country. With the lack of advanced methodologies and technology, the process of climate change tracking has become extremely expensive and suffered inaccuracies due to a lack of proper knowledge of analyzing collected data, and the lack of specific accurate hardware. Throughout this research, with the use of the MQ-135 and DHT11 sensors, ESP8266 collects carbon dioxide gas and temperature/humidity respectively and other component include a push button for detecting the current season. ESP8266 is programmed to send data over MQTT protocol, which uses Wi-Fi capability to send data to MQTT Broker. Using the MQTT protocol's Publish/Subscribe criteria, node-red subscribes to the topics defined in the MQTT broker to obtain data, which is then sent to MongoDB for permanent storage and also fed into the machine learning model for climate change/warming prediction. Different algorithms are used to evaluate this model. As result, Random Forest classifier approves itself to be the best model in evaluating the built model. This study shows that the increase in carbon dioxide gas leads to the gradual increase in the environmental temperature. Finally, the prediction clarifies that if no measures are taken presently, the climate change in Rwanda's Industrial zone will be dominated by warming periods in the future.

CCS CONCEPTS

• **Computing methodologies** → Machine learning.

KEYWORDS

Internet of Things, MQTT, MongoDB, NodeRed, Climate change prediction

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1 INTRODUCTION

Weather forecasting involves predicting the weather conditions for a specified region or area. More importantly, accurate climate forecasting is required for day-to-day activities. Weather forecasts are important to both living and nonliving things [2]. Continued research in the areas of the Internet of things (IoT) and machine learning has allowed researchers to move on to various weather forecast models. Climate change is a major problem for people all around the world. However, the Internet of Things (IoT) provides a variety of resources and technologies to assist businesses and governments in reducing the negative impact of human activities on the environment [3]. The Internet of Things has the potential to help us decarbonize our energy system, deliver modern energy systems to every human being, manage our infrastructure, and adapt to and solve climate change [4]. IoT advancements helped companies, governments, and people to mitigate climate change without losing convenience. More consumers are expected to participate in activities that allow businesses and governments to collect and analyze Big Data for environmental purposes. Connected devices pave the way for the future of climate change and environmental health [4]. In Rwanda, the problem of accurately forecasting the weather persists. Meteo Rwanda's current method is the weather station model, which involves the installation of many static weather stations around the country. Massive observatories with powerful equipment (radar, satellite, etc.) [2] now do the work, although they are limited by technological advancements and are manufacturer-dependent; nonetheless, because of the use of manual strategies and the immobility of such stations, there is no real-time climate change information and the method of predicting climate change has low accuracy and is extremely-expensive. The Internet of Things (IoT) and models based on machine learning algorithms have inspired us to work on this project because they will address the various difficulties associated with collecting and analyzing pertinent big data, reduce the cost of providing real-time climate change information, and enable useful and accurate predictions of future climate change. The goal of this project is to monitor and forecast climate change in any environment using environmental parameters including carbon dioxide, temperature, and humidity. With the help of this procedure, it is possible to take preventive action before any harm is done, following various ways to lower CO2 emissions in

the environment and achieving the country's GDP by raising clean productivity and creating clean and safe environments.

1.1 Contribution

This system is a portable real-time IoT weather forecast system that is built using very low-cost components, with ease of implementation. The contribution is based on the creation of an IoT weather station that would be installed in Rwanda Industrial Economical Zone as the project case study to collect big data while saving costs and improving accuracy as well as implementing a climate change prediction model that focuses on CO₂ levels, temperature, and humidity and providing real-time climate change information to the Rwanda population.

2 RELATED WORK

Various weather forecast models have risen as a result of ongoing research in the disciplines of the Internet of Things and Machine Learning. However, the problem of accurately forecasting or predicting the weather persists. Many research studies are conducted to improve the performance of prediction algorithms. A commonly used approach is to use several prediction algorithms in combination to get better accuracy. M.K. Nallakuruppan et al. conducted the study based on comprehending the weather prediction inconsistencies and in-proficiency in light of linear regression algorithms and time series models [2]. The significant commitment of this research is to formulate a productive weather prediction model based on a Decision Tree and Time Series Analysis. This study did not mention the approach that would be utilized to collect big data to get the most accurate results. The Applicability of Big Data in Climate Change research presented an overview of the inter-relationship between data science and climate studies, as well as a description of how Big Data tools may be used to manage sustainable climate challenges. This research aimed to highlight the perspective of systems of systems (SoS) as the drivers and effects of climate as well as that their resilience and adaptation cannot be determined without the exploration of the synergies between new research trends and disciplines. It also discussed Big Data analytics technologies and their contributions to understanding the features of climate change, as well as climate action-related equivalents such as sustainability and social sciences, which are critical for the effective development and implementation of strategies [6]. IoT based climate prediction using ANN for green networking is a proposed topology formulated an IoT-based weather monitoring using artificial neural network (ANN) for prediction. The main objective of this designed system is monitoring weather parameters such as temperature/humidity/ pressure/ rainfall etc and employing ANN for forecasting the occurrence of a flood using temperature and rainfall data using LM and GD learning algorithms [7]. Israr Ullah et al. presented a general conceptual model for learning and improving the accuracy of prediction algorithms in dynamic conditions. The proposed model evaluation and experimental analysis are conducted in a Greenhouse environment to accurately predict indoor climate conditions from noisy sensor readings and the ANN algorithm is used to improve the 8 performance of the conventional Kalman filter algorithm in dynamically changing external conditions. This study suggested that in the future, deep learning

algorithms (rather than ANN) be used to tune the performance of other prediction algorithms and that experimental analyses with big data be conducted in more complex real-world applications to further establish the validity of the proposed learning to prediction mechanism[8]. The study, titled "An Intelligent Weather Prediction System based on IoT" proposed an improved solution for weather monitoring using low-cost GPS-enabled IoT devices, connected with different sensors. The sensed information is stored in the server to predict the weather parameters such as temperature, humidity, air pressure, etc. of any defined application region of interest. This prediction can take place from the collected information to the predicted data set utilizing some soft computing tools like SVM, KNN, DNN, Linear Regression, etc. [9]. An integrated approach for estimating greenhouse gas emissions from 100 U.S. metropolitan areas found that Cities have become key players in climate change mitigation policy. To develop their climate policies, cities need good assessments of their current and future emissions. This study used publically available national datasets to develop an integrated approach for estimating GHG 4 emissions at the metropolitan level over time, between multiple locations, and across sectors [10]. According to the findings of this study, there is a lack of an appropriate method to address greenhouse gas emissions as well as sufficient large-scale data to measure progress. Tsungnan Chou et al. presented the study with the intention of extracting useful features from cardholder data and applying artificial intelligence techniques to assist financial institutions in reducing their credit card default risk. The cardholder's personal information as well as repayment details were used to track consumer behavior for the predictive models. This study also aims to compare the prediction accuracy of newly developed deep learning methods to that of conventional machine learning based on credit card holders' financial information. The experiment results showed that the deep neural network outperformed the machine learning models in most evaluation metrics and achieved an impressively high accuracy of 0.93 [23]. Priya Singh et al. conducted research with the goal of developing an efficient and accurate system to inspect cardiovascular diseases, and the system is based on data mining techniques that can help to remedy such a situation. The system is built using classification algorithms such as Random Forest, Logistic Regression, Naive Bayes' and Support Vector Machine, as well as feature selection algorithms such as Pearson Correlation and Chi-Square, to improve classification system accuracy and reduce execution time. As a result, it was discovered that Logistic Regression had the highest accuracy of 84% when compared to the others [24]. Rwanda is vulnerable to climate change as a whole. Climate change, if it occurs as predicted, would worsen Rwanda's current environmental, social, and economic concerns. Based on evaluations of 19 Global Circulation Models (GCMs) and 15 GCMs, Rwanda as a whole is anticipated to see a 0-10% rise in mean precipitation and a 2-4 degrees Celsius increase in mean temperature until 2080. The uncertainties are substantial, and they are projected to differ widely across the country [11].

3 PROBLEM DESCRIPTION

Meteo Rwanda does not monitor environment with regards to climate change. Only weather monitoring is made considering only

minimum temperature and maximum temperature of the environment. The data collected cannot fully contribute in climate change monitoring. There are no low-cost sophisticated methods or technologies for collecting and analyzing large amounts of data in order to effectively predict the weather. Climate change is the term that scientists are currently using to describe the complex changes caused by greenhouse gas concentrations that may already be impacting the planet's weather and initiating global warming and climate systems [3]. Carbon dioxide is the main greenhouse gas 4 produced by burning organic materials such as coal, oil, gas, wood, and solid waste[4]. Thus, predicting the level of CO₂ in the air is important to make decisions based on analyses to decrease air pollution. Agriculture is one of the fields that is hugely affected by climate change, and the quality and quantity of the yield are negatively impacted too. Increased animal mortality rates due to fauna and flora destruction. As a result, it's critical to fix those problems with the help of this system" IoT-based climate change prediction system", which will also be a Rwandan IoT project. This project will address those challenges of collecting and analyzing relevant data from various areas using appropriate methods and technologies and minimizing costs to provide real-time climate change information while enabling useful and accurate predictions of future climate change with a focus on temperature and carbon dioxide (CO₂) levels in Rwanda industrial economic zone.

4 METHODOLOGY

The methodology of this study aims to deploy a smart solution built upon state-of-the-art technology that uses sensors to monitor the environment and relies on Machine learning models to predict change. Data are collected by using different methods such as Observation, documents and Internet. According to data collected from different areas, it would aid in the design and implementation of a system for effective climate change monitoring and prediction.

4.1 SYSTEM ARCHITECTURE

This system architecture is drawn in figure 1. The sensing component of the data collection system under consideration in this study includes three sensors for measuring ambient temperature, relative humidity, and carbon dioxide. Other components include a push button for detecting the current season and a power source to generate the needed power in the circuit, as well as NodeMCU (ESP8266), a microcontroller platform with Internet connection capability for transmitting processed sensor data to the IoT cloud via existing Internetbased network connectivity. The dataset in this instance is created and analyzed using a machine-learning model to make predictions. A mobile and web application is also created to help a variety of users access real-time climate change data from cloud storage as well as the prediction outcomes whenever they want using any user interface device (phone, machine, etc.). The system is made up of the following components, which include both hardware and software:

NodeMCU(ESP8266): is a WiFi-capable microcontroller development board. It employs an ESP8266 microcontroller chip, as well as an integrated Wi-Fi communication protocol and multiple input/output pins.

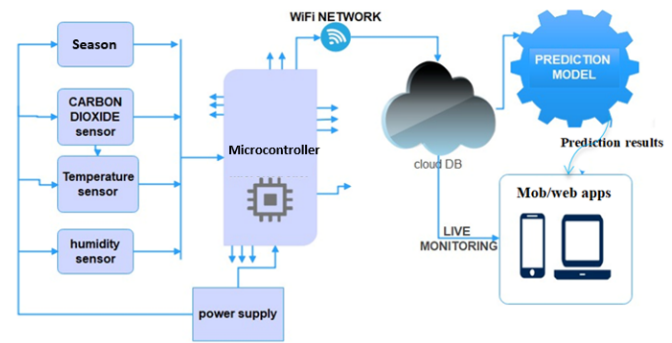


Figure 1: General Architecture

MQ-135: is a popular gas sensor from the MQ series that is commonly used in air quality control equipment. It can detect various gases, including CO₂, and operates between 2.5V and 5.0V, consuming approximately 150mA [18]. It requires some pre-heating before it can produce accurate results, and it can output both digital and analog signals.

DHT11 sensor: is a basic, low-cost digital temperature and humidity sensor. It measures the surrounding air with a capacitive humidity sensor and a thermistor and outputs a digital signal on the data pin [22].

Digital push button: is employed to give a triggering pulse that sends a 1 to the microcontroller when the climate is warm and sends 0 when the climate is cold [19].

Python: is a general-purpose programming language that is open to the public and is regarded as a high-level coding language. Python is an interpretive, easy-to-use coding language that uses object-orientation concepts and is considered as a beginner level coding language [21].

MQTT Broker: is employed as a cloud server that holds payload temporarily to redirect it to different far ends and gives the data once a client subscribes to it[12][15]. MQTT Broker is built upon MQTT Protocol, a lightweight machine-to-machine communication protocol that transport messages from one node to another, with a tiny bandwidth utilization and code overhead.

Node-RED: is a development tool for tying new and intriguing connections between physical components, APIs, and web services. With the help of the extensive selection of nodes in the palette that can be deployed to its runtime with a single click, it offers a browser-based editor that makes it simple to wire up flows[13]. This tool shows the deployed nodes that briefly include the mqtt-in node, nodes of functions to read payload from topics, messages debugger, gauges and charts for data visualization, and mongodb2 in and out nodes.

MongoDB: is a No-SQL database that stores data in collections. With python script, the data is retrieved from the Mongo database to be fed to the ML, which continuously studies under the train-test-split function to give an accurate prediction/forecast. MongoDB as a no-SQL, saves data in a different format. Each row is called a document, and each document has a corresponding ID. The collection of documents forms a collection [14][20]. In this study,

The saved data is of Temperature, Humidity, Carbon dioxide, and Season.

5 IMPLEMENTATION AND RESULT

5.1 System Design

As was previously detailed, the system is discussed in two parts, namely: Hardware part made of sensors, push button, microcontroller, power source, and cables and Software part made of Arduino IDE, MQTTBox, Node-red, MongoDB, and Anaconda based Jupyter notebook for python processing. The following figure 2 summarizes the entire system design of the "IoT-based Climate Change Prediction System."

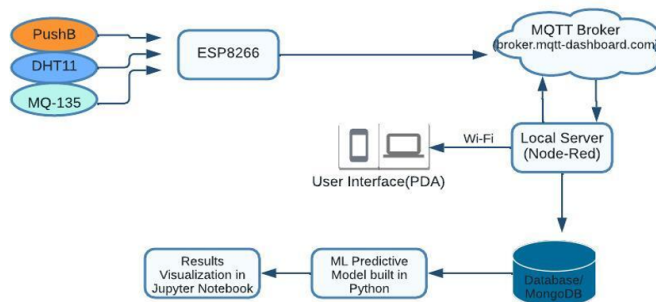


Figure 2: System Design

According to the figure 2, the collected data will be published at the given topic to MQTT Broker via IoT Gateway specifically network router or access point which is connected directly to internet connectivity. At the other side of the MQTT broker, the client server(local server) was implemented by using Node-Red software tool, subscribes to the same topic and MQTT Broker as well as did by the publishing field sensor nodes(ground weather station). The received data by the Node-Red data collector nodes by the help of MQTT IoT communication protocol are visualized to the user interface and pushed into the database (MongoDB) to be stored permanently for later use.

Lastly, through python application software tools the data in the database are collected into the format needed to be used by the Machine Learning predictive model generated by model training. Finally, the data from the database are combined with the ML predictive model to generate new predictions and the resulted prediction is visualized in Jupyter Notebook.

5.2 Data collection in Arduino

The Arduino codes are written in c/c++, it is a sample taken from Arduino IDE to show how the MQTT server is defined and accessed, by using a call of "broker.mqtt-dashboard". MQTT port number is 1883 as shown in the lines below.

```

louis_thingspeak | Arduino 1.8.16
File Edit Sketch Tools Help
Upload
louis_thingspeak

const char* mqtt_server = "broker.mqtt-dashboard.com"; // MQTT Broker
int mqtt_port = 1883;
WiFiClient espClient;
PubSubClient mqttclient(espClient);
long lastMsg = 0;
char msg[50];
int carbondioxide = 0;
int humidity = 0;
int temperature = 0;

void setup() {
  Serial.begin(115200);
  // Start up the library
  setup_wifi();
  mqttclient.setServer(mqtt_server, mqtt_port);
  mqttclient.setCallback(callback);

  Serial.println("Connected ");
  Serial.print("MQTT Server ");
  Serial.print(mqtt_server);
  Serial.print(":");
}


```

The others set of codes shows how to define at which topics the MQTT client publishes the payload as shown in the below lines.

Once the codes are successfully compiled and uploaded to the ESP8266, the serial monitor output the results in showing the status of the Wi-Fi connection, the corresponding IP address, and once the mqtt server is connected, displays the readings from the sensors.

```

carbondioxide= analogRead(A0UTpin);
char carbondioxideString[8];
dtostrf(carbondioxide, 1, 2, carbondioxideString);
Serial.print("carbondioxide: ");
Serial.println(carbondioxideString);
mqttclient.publish("aceiot/louise/carbondioxide", carbondioxideString);

humidity = dht.readHumidity();
// Read Humidity as Celsius (the default)
char humString[8];
dtostrf(humidity, 1, 2, humString);
Serial.print("Humidity: ");
Serial.println(humString);
mqttclient.publish("aceiot/louise/humidity", humString);

```

5.3 Model Building

Throughout this research project, a model is built upon Keras. The dataset used contains 521 samples for four different variables, including environmental temperature, relative humidity, carbon dioxide, and seasonal data that show the precise weather conditions of data collection. The dataset was initially split into two smaller datasets, a training dataset and a testing dataset. 25% of the original dataset is in the testing dataset, while 75% is in the training dataset. The training and testing processes were implemented through python programming codes and python libraries found in sklearn python package. These processes were started by importing the original dataset from CSV file by using dataframe implemented with panda's python module.

Once the model is trained and tested, it is evaluated. Evaluating a model is the process of passing a model through other pre-built models to increase its training and testing accuracy. The selection of the specific evaluation metrics depends on the type of category of machine learning problem. The metrics techniques used during of model evaluation in this research study includes Accuracy, Recall,

Precision [17]. These evaluation metrics are defined using confusion matrix and were selected because they are the best fit for machine learning classification problems. The confusion matrix is one of machine learning techniques for summarizing the performance of a machine learning classification models. As shown in the figure 3, the matrix contains four main elements that include True Positive (TP), False Positive (FP), True Negative and False Negative[25].

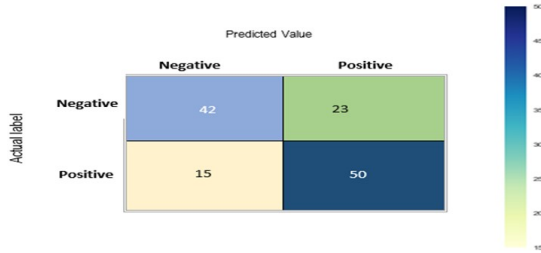


Figure 3: Model study with Confusion Matrix

Once the model is built, it is evaluated as illustrated in figure 4. The training accuracy of the built model is 79% with the corresponding testing accuracy of 78%.

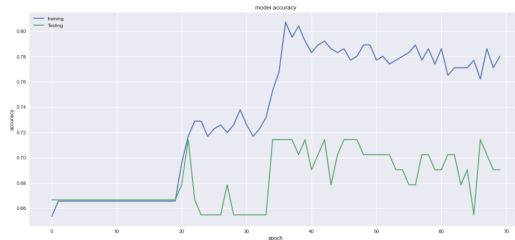


Figure 4: Train-Test Model Evaluation

5.4 Training and evaluation implementation

This study is generally a classification study because, in the end, a forecast tells whether the climate is warm or cold. The dataset for machine learning algorithms was prepared, and then the models were trained on training data using various machine learning classification models including Logistic regression (LR), Gradient Boost Classifier (GB), Random Forest Classifier (RF), Multilayer Perceptron classifier (MLP), Decision Tree Classifier (DT), K Nearest Neighbor Classifier (KNN) and Facebook Prophet Predictive model. These machine learning algorithms were used for training and testing data as well as for mapping the relationships between the dependent variables (model input) and independent variables (model output). Each model performs differently from others and gives its independent training and testing accuracy scores.

5.4.1 Random Forest classifier. The model built has a training accuracy of 79% and a testing accuracy of 78%. Trying to increase the accuracy by evaluating the model by using the Random Forest classifier model, the Training accuracy rises to 99% with the corresponding testing accuracy of 84% as figure5 illustrates, and

both accuracies continue to increase as the number of estimators increase.

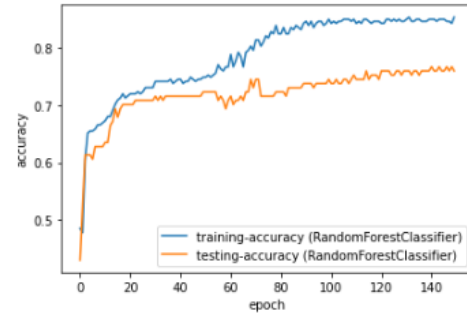


Figure 5: Model evaluation with Random Forest

5.4.2 Multilayer Perceptron Classifier. Once the model is evaluated by using Multilayer Perceptron, the accuracy improves as follows: both training and testing accuracies are 78% as illustrated in figure 6, and both accuracies continue to increase as the number of iterations increases.

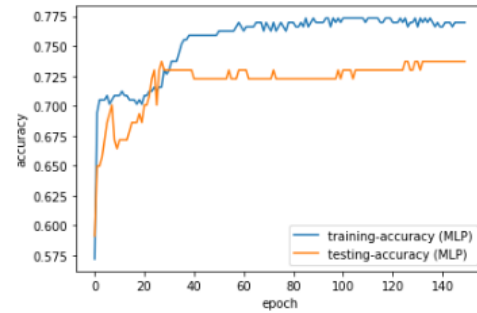


Figure 6: Model evaluation with Multilayer Perceptron

5.4.3 Logistic Regression. Employing Logistic Regression, the training accuracy rises to 82% with the corresponding testing accuracy of 80% as figure 7 shows, and both accuracies continue to increase as the number of iterations increases.

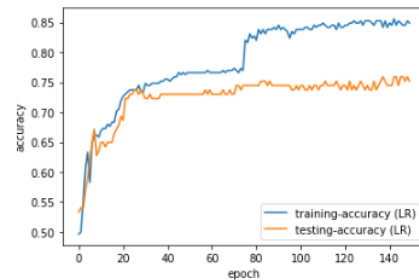


Figure 7: Model evaluation with Logistic Regression

5.4.4 Gradient Boost Classifier. Evaluating the built model with the Gradient Boost, the training accuracy rise to 97% with a corresponding 77% as figure 8 illustrates, and both accuracies continue to increase as the number of estimators increase.

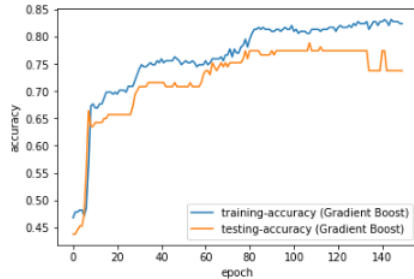


Figure 8: Model evaluation with Gradient Boost Classifier

5.4.5 Decision Tree Classifier. When the model is evaluated by employing the Decision Tree Classifier model, the accuracies become as follows: Training accuracy of 99% and 79% of testing accuracy as figure 9 shows. And both accuracies continue to increase as the maximum depth increase.

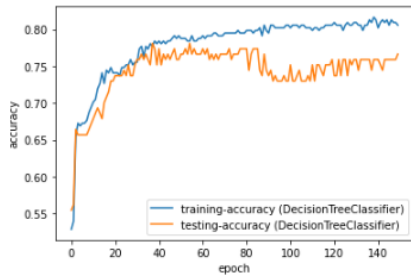


Figure 9: Model evaluation with Decision Tree Classifier

5.4.6 K Nearest Neighbor Classifier. When aggregating the built model with K Nearest Neighbor Classifier, the training accuracy rises to 84 % and the testing accuracy to 77% as illustrated in figure 10, and both accuracies continue to increase as the number of neighbors increases.

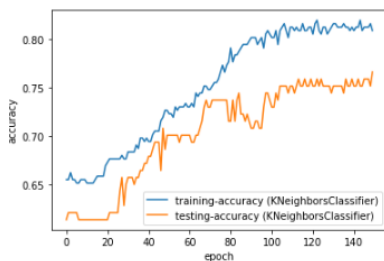


Figure 10: evaluation with K Nearest Neighbor

5.5 Comparative Analysis

When compared to other machine learning algorithms, Random Forest has high prediction accuracy. This indicates that this system will perform better with Random Forest than with other machine learning algorithms, as shown in table 1.

Table 1: Comparative analysis with different algorithms

ALGORITHM	ACCURACY	
	TRAINING	TESTING
Logistic Regression (LR)	82%	80%
Gradient Boost Classifier (GB)	97%	77%
Random Forest Classifier (RF)	99%	84%
Multilayer Perceptron classifier (MLP)	78%	78%
Decision Tree Classifier (DT)	99%	79%
K Nearest Neighbor Classifier (KNN)	84 %	77%

5.6 Prediction with Prophet Facebook Model

The prophet model is built by Facebook META. The prophet is utilized in numerous Facebook applications to generate accurate forecasts for planning and goal-setting, it can predict the future based on past and current data [16]. With this model, when the same dataset is fed to it to forecast the warming of the coming year, figure 11 shows that there will be a continuous increase of the warming periods throughout the year 2023 in our country. As the season is Warm or Cold, the measurements below Zero and the measurements above one are not considered during the interpretation of the graph. The general meaning of the illustrations is to let observers see that there will be a gradual increase in the periods of warming throughout the year.

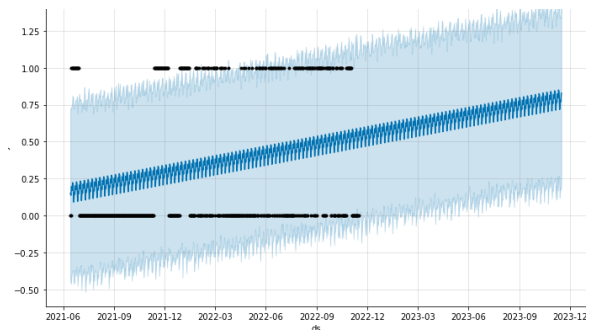


Figure 11: Prophet Model-Based Prediction

5.7 Live Data Visualization

The live data visualization from sensors by using Node-red is illustrated below in figure 12, Carbon Dioxide gas CO₂ is ranging from zero to 300 PPM, Humidity is ranging from 0 to 100%RH, and Temperature is ranging from -10 to 50 °C.

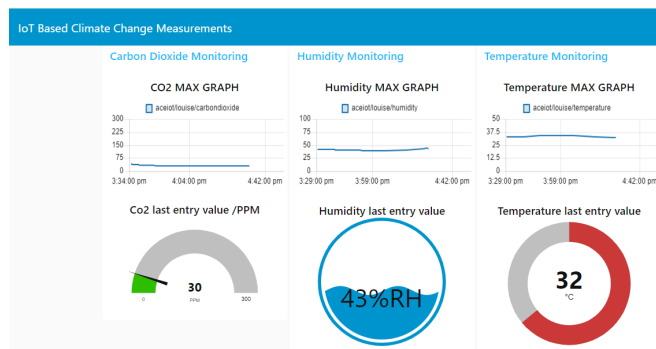


Figure 12: Live Data Visualization in charts and gauges

6 CONCLUSION AND FUTURE WORK

During this study, we studied the use of the state-of-the-art technologies of IoT and Machine learning and deep learning aspects to predict climate change in Rwanda Industrial Economic Zone. The model used is built by using Jupyter notebook which is a good tool for python users. This model is trained and evaluated by using different algorithms that include Logistic Regression (LR), Multi-Layer Perceptron (MLP), Gradient Boost (GB), K Nearest Neighbor (KNN), Decision Tree Classifier (DT), and Random Forest Classifier. Random Forest approves itself to be the best model in evaluating the built model with 99% of training accuracy and 84% of testing accuracy. The study shows that once the carbon dioxide gas emission goes higher, the temperature in the environment increases, which directly or indirectly affects the environment in general. Finally, the prediction clarifies that if no measures are taken presently, the climate change in Rwanda's industrial zone will be dominated by warming periods in the future. It is proved. The employed technologies are fully trustworthy and we can recommend them to other technologists. We are recommending to future researchers have enough hardware to collect data from different parts of the country throughout different periods of the year to predict the warming of the wide environment also, we are recommending them have a bit larger dataset so that the model to build becomes more accurate.

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